Internal Assessment – I
Advanced Power Electronic Converters IA Project
Even Semester 2026

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Title of the Project:	**Design and Harmonic Analysis of a Fault-Tolerant 5-Level CHB Inverter using LSPWM-PD for EV Traction**

Type (Select any one):

Simulation	
Hardware	
Simulation and Hardware	

Mention the title of the topic if you have applied for SRP:	NA

1.1 ABSTRACT :

- This project presents a 5-level CHB inverter **[1]** for an EV traction motor **[2]**, controlled by Level-Shifted PWM.
- An LC filter is used to reduce the normal operating THD to 2.24%, strictly passing IEEE-519 standards.
- A "limp-home" logic is designed to bypass any failed hardware switch, ensuring the car keeps moving instead of stalling.

1.2 OBJECTIVES :

- Design a clean power drive: Build the inverter and tune the LC filter to keep total harmonic distortion under 5%.
- Create a safety backup: Implement fault-tolerant logic that isolates a broken cell and prevents a total system shutdown.
- Compare harmonic data: Analyze the FFT graphs **[3]** to see how the system behaves during raw, filtered, and faulted states

1.3 WORK CARRIED OUT :

- **The Main Build:** We built a 5-level Cascaded H-Bridge inverter in Simulink, specifically set up to run an electric vehicle motor.

- **Custom Coding:** Instead of using standard pre-built blocks, we wrote the logic for the Level-Shifted PWM entirely from scratch.

- **Meeting Real Standards:** We spent a lot of time tuning an LC filter to clean up the power. We got the distortion (THD) down to just 2.24%, which officially passes the strict IEEE-519 industry limit. **[4]**

- **The "Limp-Home" Feature:** This is the core of the project. If a hardware switch breaks, the EV doesn't just die on the road. Our logic catches the fault, cuts out the broken part, and drops to a 3-level output so the car can safely "limp home." **[5]**

- **Tracking the Breakdowns:** We ran FFT graphs to show exactly what happens to the power during two stages: Healthy State, and Faulted state, both before and after the LC filter is applied.

- **Proving It Stays Safe:** We created an XY graph (Voltage vs. Current) that proves our design is safe. It shows that when the car suddenly drops from normal driving into the emergency limp-home mode, the motor stays completely stable and doesn't lose control.

- **Smart Hardware Setup:** We achieved the 5-level power output using 8 main switches (IGBTs). It is a highly efficient way to get clean power to the motor without making the physical hardware unnecessarily huge or complicated.

- **Fixing the Math Glitches:** High-speed switching (4000 Hz) causes a lot of calculation errors in standard software. We had to specifically fine-tune the simulation solvers and the modulation index (setting it exactly to 0.95) to stop the waveform from glitching out at the peak voltages.

- **Smooth Shifting:** When the hardware breaks and drops to the 3-level backup mode, the transition isn't violent or jerky. The graphs prove that the power shifts down smoothly, which prevents the motor from taking extra damage during the emergency.

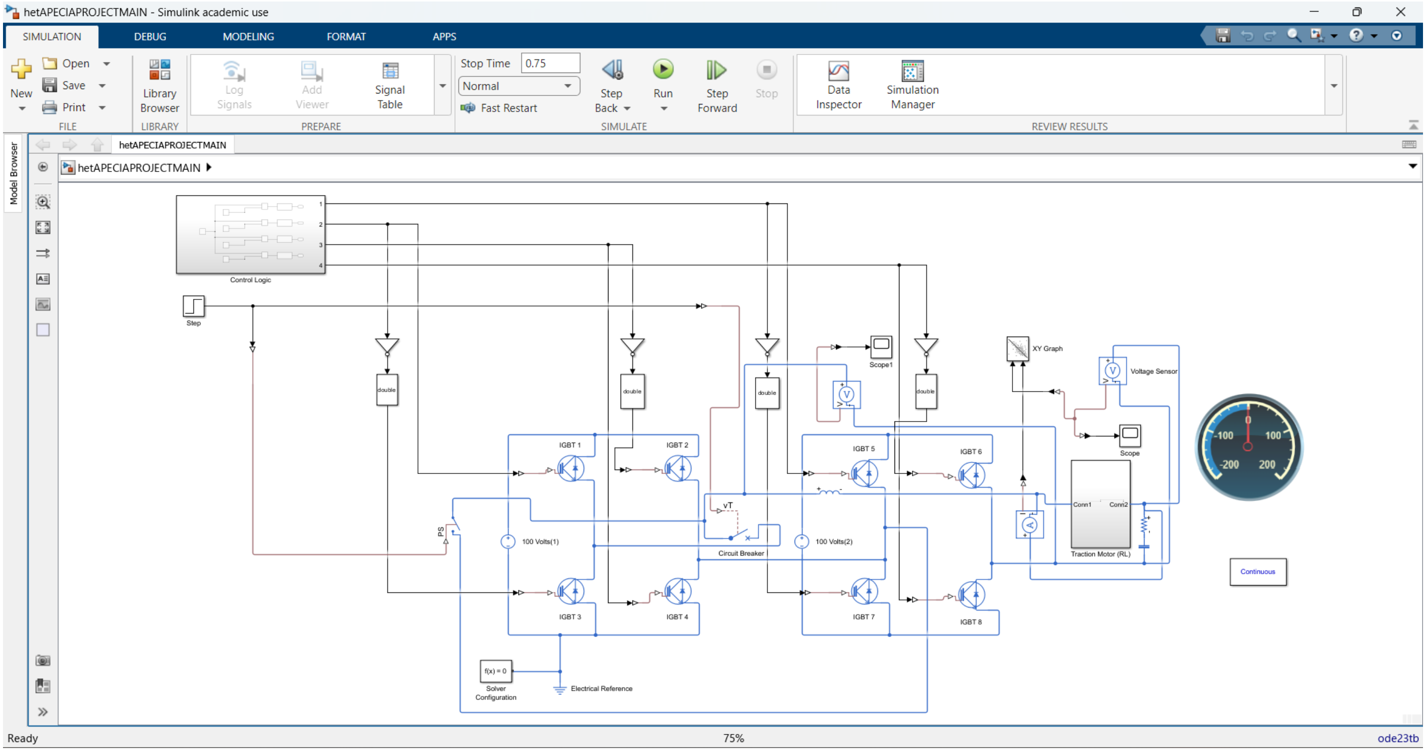
- **Tuning Out the Glitches:** During testing, we noticed the power was dropping out and creating gaps at the extreme negative voltages. We had to manually debug the math behind the high-speed switching and found the perfect "spot" (setting the amplitude to exactly 1.9) to make the final power output perfectly solid and reliable

- **FFT Analyzer:** We calculated the THD using the Powergui block for both Healthy and Faulted conditions. We found that the THD remained very low (2.24%) after the Filter was applied, but almost as high as 30.59% when filter is not applied, thus concluding that Filters are very useful in such conditions.

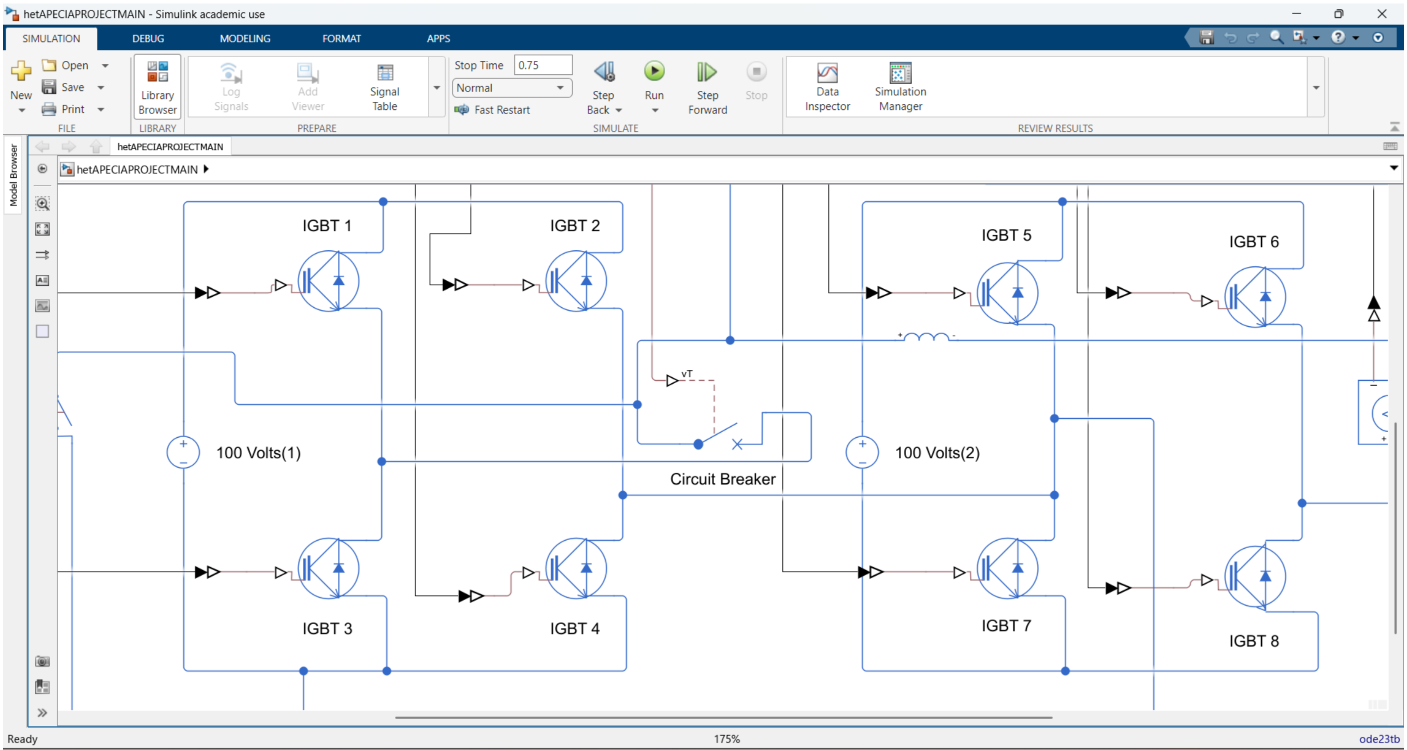
- **Platform:** MATLAB/SIMULINK R2024b version was the platform utilized and the Variable Step Discrete solver – ode23tb was selected for this specific project. **[6]**

1.4 RESULTS :

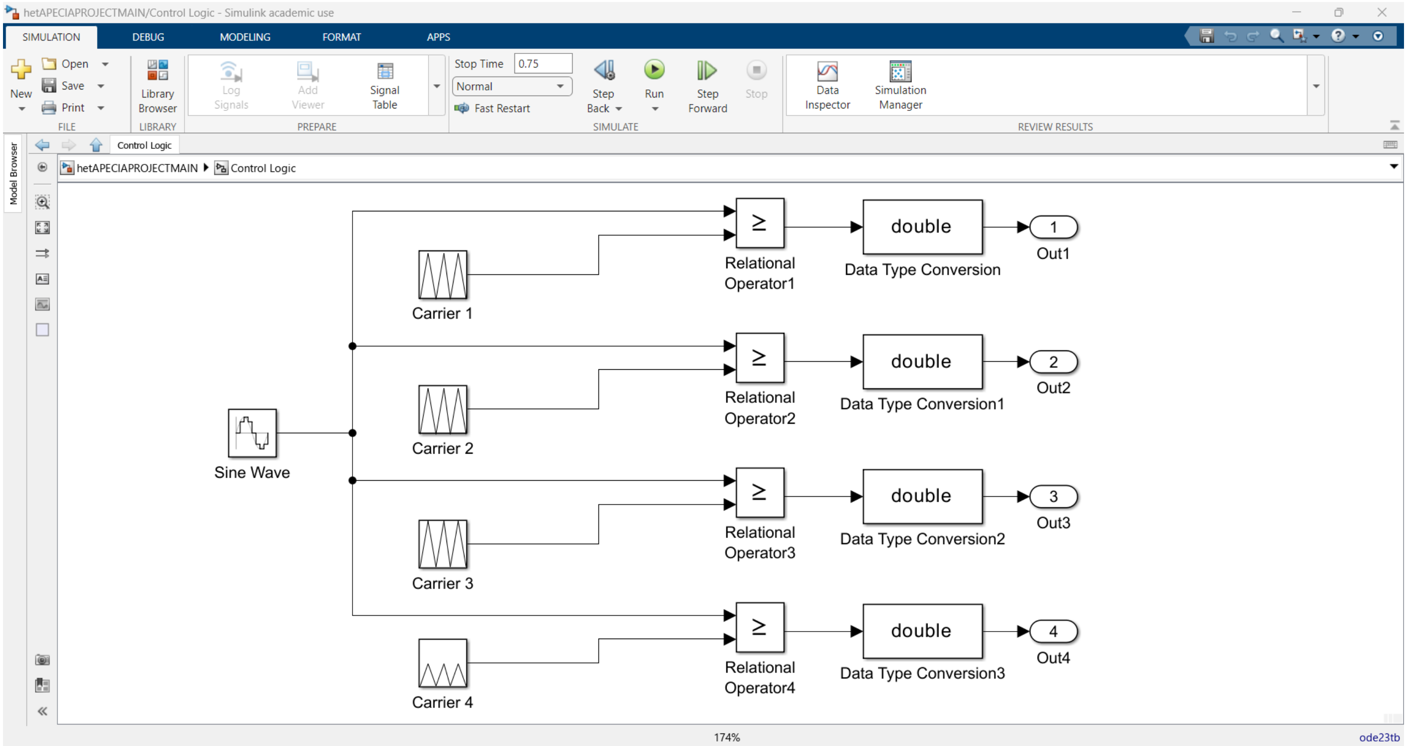
SIMULINK MODEL IMAGE : CASCADED H – BRIDGE 5 LEVEL INVERTER [7]



CLOSE – UP VIEW OF BOTH THE H – BRIDGES (8 IGBTS, 4 LEGS) :



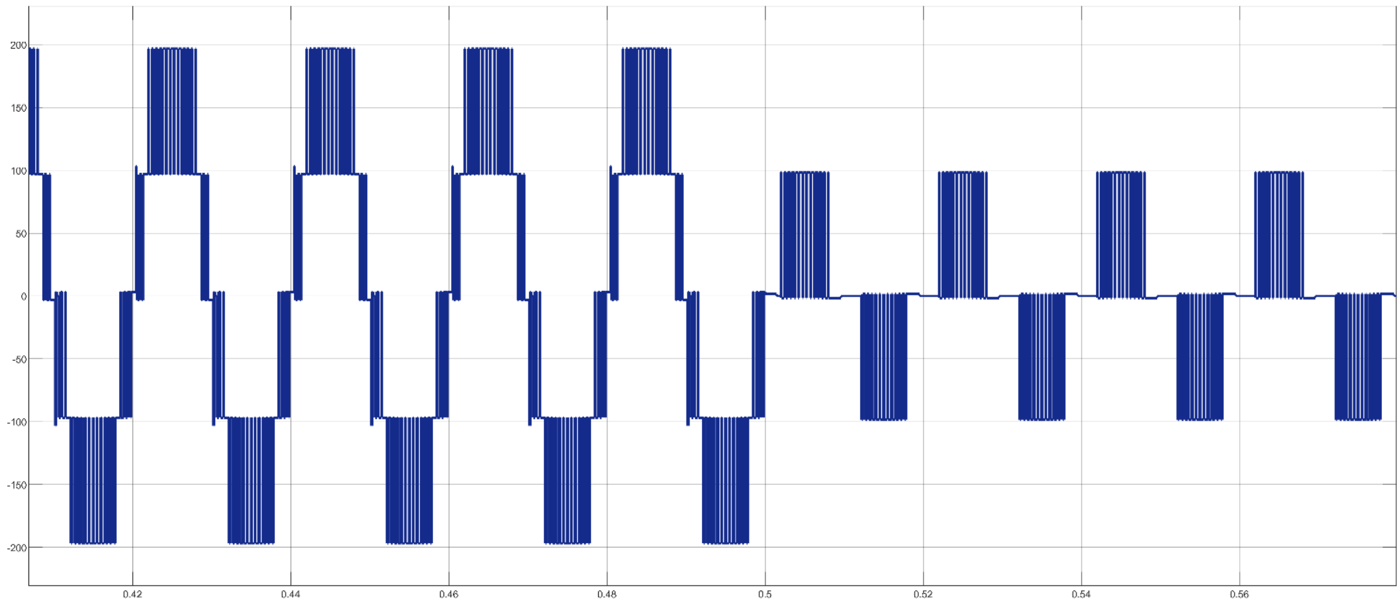
CONTROL LOGIC OF THE CASCADED H BRIDGE INVERTER : (LSPWM – PD) [8]
(LEVEL SHIFTED PULSE WIDTH MODULATION – PHASE DISPOSITION)



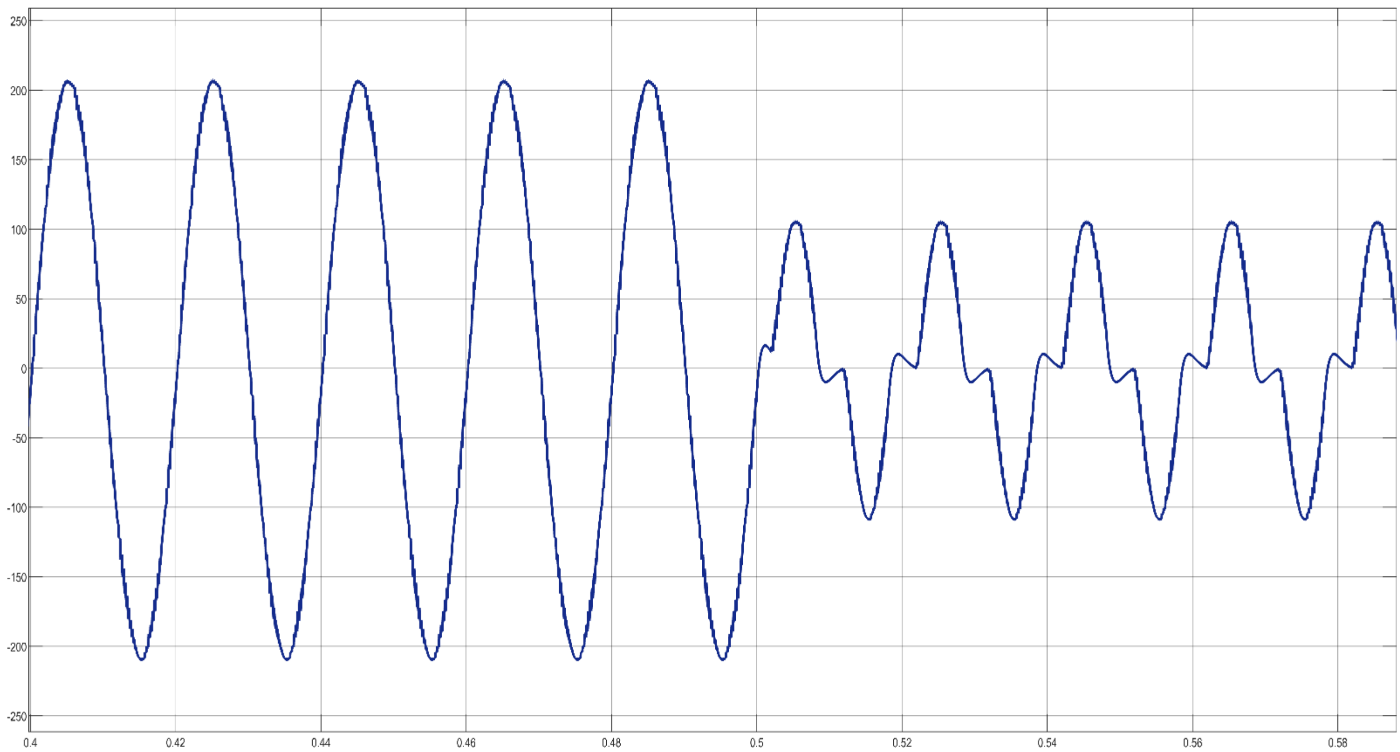
CONTROL LOGIC SPECIFICATIONS TABLE :

Block Name / Subsystem	**Parameter**	**Value / Setting**
Sine Wave (Reference Signal)	Amplitude	1.90
	Frequency	2*pi*50
	Phase	0
	Sample Time	0.0001 or 1e-4
Carrier Waves (Repeating Sequence)	Common Time Values (All Carriers)	[0, 0.000125, 0.00025] seconds
	Carrier Frequency	4000 Hz
	Carrier 1 Output Values	[1, 2, 1]
	Carrier 2 Output Values	[0, 1, 0]
	Carrier 3 Output Values	[-1, 0, -1]
	Carrier 4 Output Values	[-2, -1, -2]
Relational Operators	Operator Type	Greater than or equal to (≥)
	Input 1 (Top)	Sine Wave
	Input 2 (Bottom)	Respective Carrier Wave
Data Type Conversion	Output Data Type	double
Simulation Environment	Stop Time	0.75 seconds or 1 second

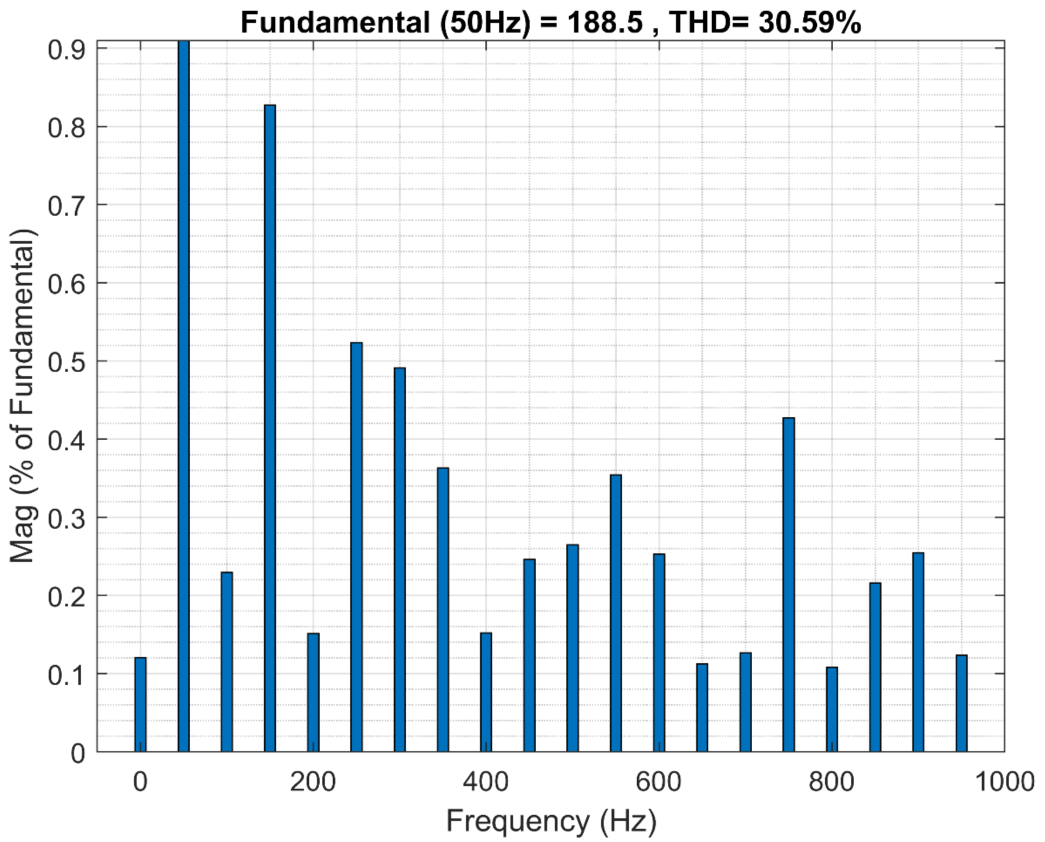
CASCADED H BRIDGE INVERTER SCOPE : 5 LEVELS : +200Vdc, +100Vdc, 0, -100Vdc, -200Vdc + SCOPE AFTER SHORT CIRCUIT FAULT AT T = 0.5 SECONDS : 3 LEVELS : +100Vdc, 0, -100Vdc :



FILTERED WAVEFORM (PURE SINE WAVE) AFTER PASSING THROUGH LC FILTER :

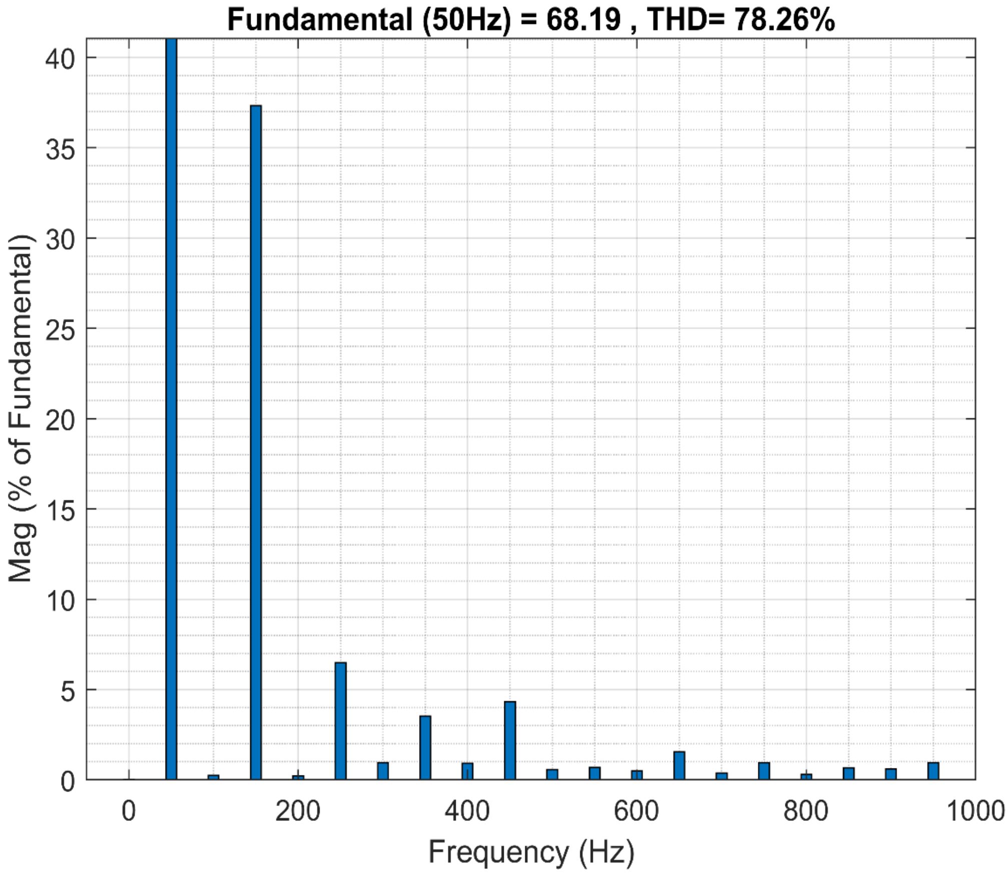


TOTAL HARMONIC DISTORTION OF RAW 5 LEVEL STAIRCASE : FUNDAMENTAL = 188.5, THD = 30.59%
(Before passing through LC filter)



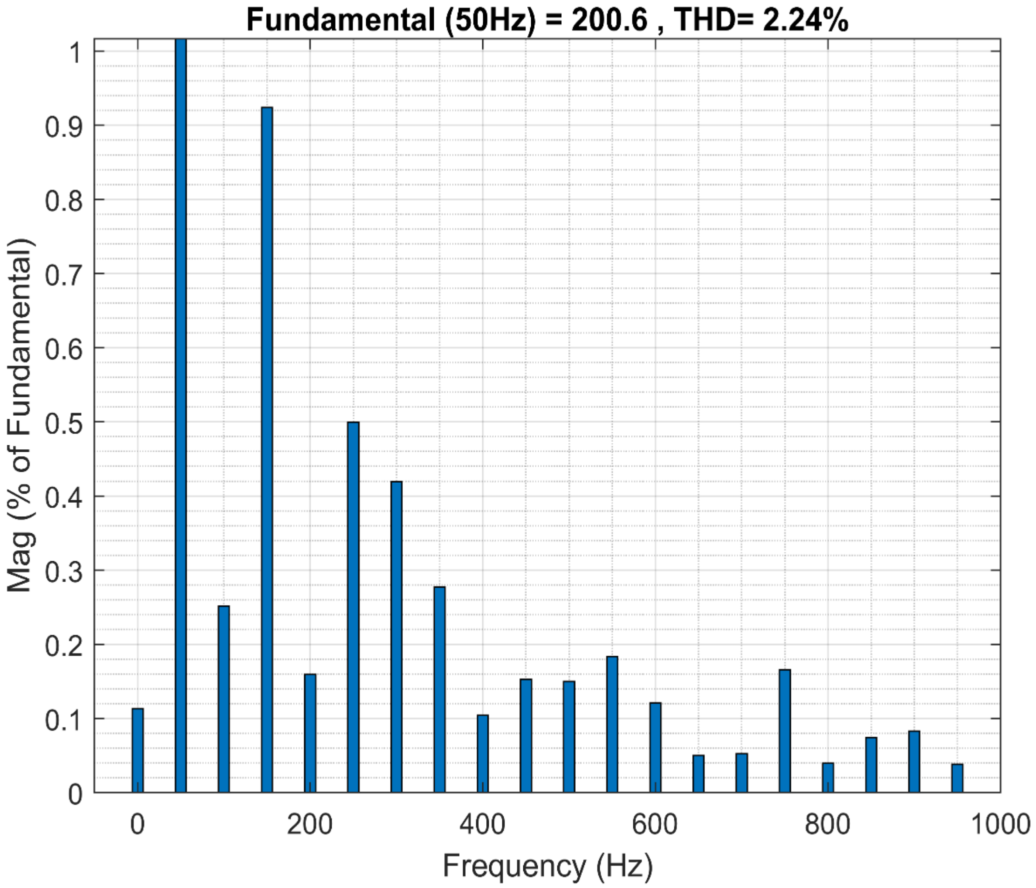
Frequency (Hz)	Harmonic Order (n)	Magnitude (% of Fundamental)
0	DC Component	~0.12
50	1 (Fundamental)	100.0 (Off-scale peak)
100	2	~0.23
150	3	~0.83
200	4	~0.15
250	5	~0.52
300	6	~0.49
350	7	~0.36
400	8	~0.15
450	9	~0.25
500	10	~0.26
550	11	~0.35
600	12	~0.25
650	13	~0.11
700	14	~0.13
750	15	~0.43
800	16	~0.11
850	17	~0.22
900	18	~0.26
950	19	~0.12

TOTAL HARMONIC DISTORTION (POST FAULT) 3 LEVELS : FUNDAMENTAL = 68.19, THD = 78.26%
(Before passing through LC filter)



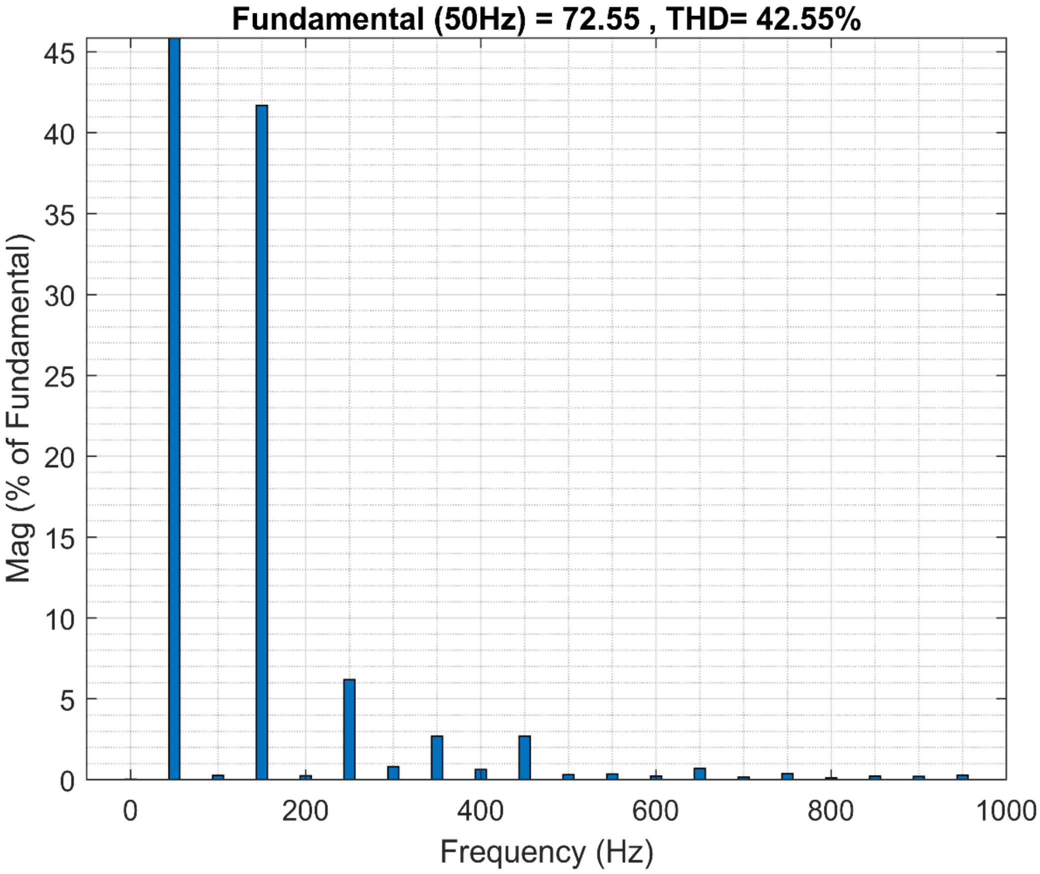
Frequency (Hz)	Harmonic Order (n)	Magnitude (% of Fundamental)
0	DC Component	~0.15
50	1 (Fundamental)	100.0 (Off-scale peak)
100	2	~0.20
150	3 (Dominant Unbalance)	~37.30
200	4	~0.20
250	5	~6.50
300	6	~0.90
350	7	~3.50
400	8	~0.90
450	9	~4.30
500	10	~0.50
550	11	~0.60
600	12	~0.40
650	13	~1.60
700	14	~0.30
750	15	~0.90
800	16	~0.30
850	17	~0.70
900	18	~0.60
950	19	~0.90

TOTAL HARMONIC DISTORTION OF FILTERED WAVEFORM : FUNDAMENTAL = 200.6, THD = 2.24%
(After passing through the LC filter)



Frequency (Hz)	Harmonic Order (n)	Magnitude (% of Fundamental)
0	DC Component	~0.11
50	1 (Fundamental)	100.0 (Off-scale peak)
100	2	~0.25
150	3	~0.92
200	4	~0.16
250	5	~0.50
300	6	~0.42
350	7	~0.28
400	8	~0.11
450	9	~0.15
500	10	~0.15
550	11	~0.18
600	12	~0.12
650	13	~0.05
700	14	~0.05
750	15	~0.17
800	16	~0.04
850	17	~0.07
900	18	~0.08
950	19	~0.04

TOTAL HARMONIC DISTORTION OF FILTERED WAVEFORM (POST FAULT)
(After passing through the LC filter), FUNDAMENTAL = 72.55, THD = 42.55%

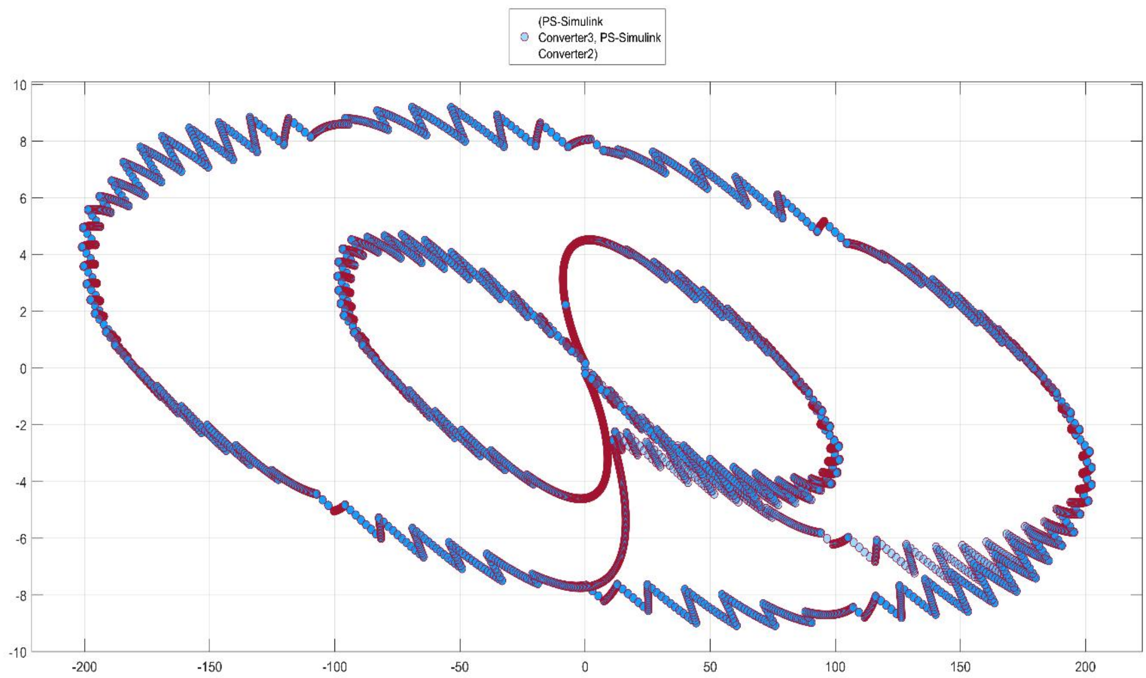


Frequency (Hz)	Harmonic Order (n)	Magnitude (% of Fundamental)
0	DC Component	~0.10
50	1 (Fundamental)	100.0 (Off-scale peak)
100	2	~0.20
150	3 (Dominant Unbalance)	~41.80
200	4	~0.20
250	5	~6.20
300	6	~0.80
350	7	~2.70
400	8	~0.60
450	9	~2.70
500	10	~0.30
550	11	~0.40
600	12	~0.30
650	13	~0.70
700	14	~0.20
750	15	~0.40
800	16	~0.10
850	17	~0.20
900	18	~0.20
950	19	~0.30

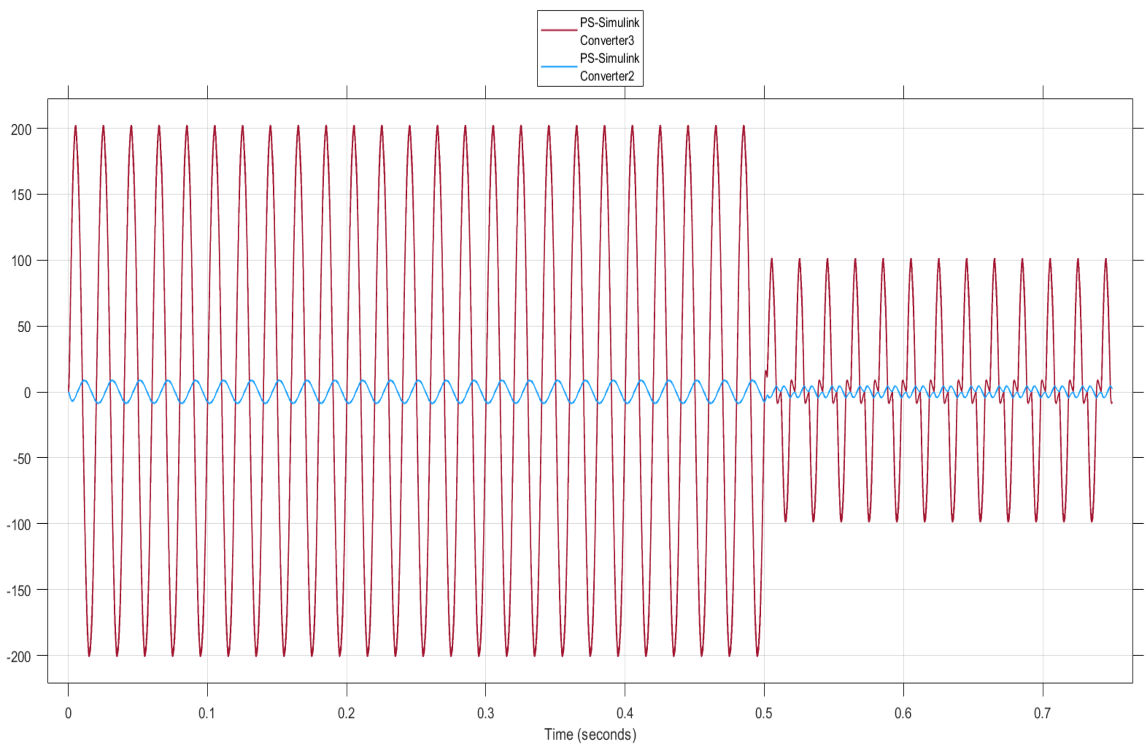
Parameter Category	Specific Parameter	Value / Unit
Software & Solver	Simulation Environment	MATLAB / SIMULINK R2024b
	Solver Configuration	Ode23tb
	Maximum Step Size	0.0001 seconds
	Simulation Stop Time	0.75 seconds
Hardware Topology	Inverter Configuration	Cascaded H-Bridge Inverter (CHBI)
	Switching Devices	8x Insulated-Gate Bipolar Transistors (IGBTs)
	Number of Output Levels	5-Level (Normal) / 3-Level (Limp-Home Fault)
Control Logic	Modulation Technique	Level-Shifted Phase Disposition (LSPWM-PD)
	Amplitude Modulation Index	0.95
	Frequency Modulation Ratio	80
	Fundamental Frequency	50 Hz
	Carrier Switching Frequency	4000 Hz
Filter Design	Filter Topology	Passive LC Low-Pass
	Filter Inductance	5 mH
	Filter Capacitance	150 μF
	Filter Resonant Frequency	~184 Hz
Load Parameters	Load Representation	Series RL (EV Traction Motor Stator)
	Stator Resistance	10 Ω
	Stator Inductance	5 mH
Performance Metrics	Unfiltered Output THD (Healthy)	30.59 %
	Unfiltered Output THD (Faulted)	78.26 %
	Filtered Output THD (Healthy)	2.24 % : IEEE-519 Rule (< 5%)
	Filtered Output THD (Faulted)	42.55 %

OVERALL SYSTEMS SPECIFICATIONS TABLE

XY GRAPH OF HEALTHY + FAULTY CONDITONS (VOLTAGE SENSOR AND CURRENT SENSOR AS X AXIS AND Y AXIS RESPECTIVELY) :



TIME PLOT GRAPH :



RED SINE WAVE : Motor Voltage, BLUE SINE WAVE : Motor Current

1.5 REFERENCES

[1] L. M. Tolbert, F. Z. Peng and T. G. Habetler, "Multilevel converters for large electric drives," in IEEE Transactions on Industry Applications, vol. 35, no. 1, pp. 36-44, Jan.-Feb. 1999.

[2] Z. Amjadi and S. S. Williamson, "Power-Electronics-Based Solutions for Plug-in Hybrid Electric Vehicle Energy Storage and Traction Systems," in IEEE Transactions on Industrial Electronics, vol. 57, no. 2, pp. 608-616, Feb. 2010.

[3] E. Oran Brigham, The Fast Fourier Transform and Its Applications. Englewood Cliffs, N.J.: Prentice-Hall, 1988.

[4] IEEE Standard 519-2022, IEEE Standard for Harmonic Control in Electric Power Systems, IEEE, 2022.

[5] P. Lezana, J. Pou, T. A. Meynard, J. Rodriguez, S. Ceballos and F. Richardeau, "Survey on Fault Operation on Multilevel Inverters," in IEEE Transactions on Industrial Electronics, vol. 57, no. 7, pp. 2207-2218, July 2010.

[6] MathWorks, MATLAB and Simulink Release 2024b, The MathWorks Inc., Natick, Massachusetts, United States, 2024.

[7] M. H. Rashid, Power Electronics Handbook, 4th ed. Oxford, U.K.: Butterworth-Heinemann, 2017.

[8] B. P. McGrath and D. G. Holmes, "Multicarrier PWM strategies for multilevel inverters," in IEEE Transactions on Industrial Electronics, vol. 49, no. 4, pp. 858-867, Aug. 2002.